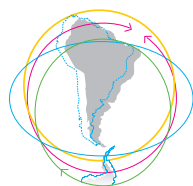




ABSTRACTS



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DROUGHT VARIABILITY ACROSS THE ANDES DURING THE LAST MILLENNIA AND ITS LINKAGES TO PACIFIC OCEAN CLIMATE VARIABILITY

Christie, D.A.¹, Barichivich, J.², Morales, M.S.³, Cook, E.R.⁴, LeQuesne, C.¹, Villalba, R.³, Alvarez, C.A.¹, Silva, J.S.¹, Muñoz, A.¹, Melvin, T.M.², Briffa, K.R.², Pacajes, J.⁵ & Samaniego, H.¹

¹ Facultad de Ciencias Forestales y Recursos Naturales, Universidad Austral de Chile, Valdivia, Chile. ² Climatic Research Unit, University of East Anglia, Norwich, UK. ³ Departamento de Dendrocronología e Historia Ambiental, Instituto Argentino de Nivología, Glaciología y Ciencias Ambientales, Mendoza, Argentina. ⁴ Lamont-Doherty Earth Observatory of Columbia University, Palisades, NY, USA. ⁵ Laboratorio de Dendrocronología e Historia Ambiental, Instituto de Investigaciones Geológicas y del Medio Ambiente, Universidad Mayor de San Andrés, La Paz, Bolivia.

duncan.christie@docentes.uach.cl

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Since the second half of the 20th century the Andes Cordillera has experienced a positive warming trend, an increase in elevation of the 0°C isotherm, rapid melting of ice caps, and a sustained shrinkage of glaciers. These climate and environmental changes have occurred in conjunction with a growing demand for hydrological resources as a result of population increases and industrial development. Given the short span of instrumental hydroclimate records for this region, longer records are needed to understand the nature of climate variability and how the dominant interannual modes of climate variability have evolved under changes in long-term background conditions. We develop a field reconstruction of summer Palmer Drought Severity Index (PDSI) since AD 1200 from 10°S to 40°S latitude based on a multi species network of tree-ring chronologies from both flanks of the Andes. Our results show two dominant spatio-temporal modes, the first centered in the Altiplano region and the second in the Central Andes of Chile and Argentina (CCha). Both modes show an intermittent anti-phase relationship among themselves at high spectral frequencies and an increase in interannual variability within the ENSO band since the late XIX century. Sea surface temperatures from the central Pacific Ocean results negatively (positive) correlated with Altiplano (CCha) summer PDSI.